Day 106 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Find Triangle Area

Standards: 6.G.A.1

Learning target: I can find the area of any triangle using a perpendicular height.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Two matching triangles form a parallelogram, so A = (base × perpendicular height)/2. A slanted side is not necessarily the height. Label area in square units.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 23

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01 / Learn

Two matching triangles form a parallelogram, so A = (base × perpendicular height)/2. A slanted side is not necessarily the height. Label area in square units.

Worked example 1

Sketch a triangle with base 4 cm and perpendicular height 5 cm. Find its area.

1. Two matching triangles form a parallelogram, so A = (base × perpendicular height)/2.

2. A slanted side is not necessarily the height.

3. A = bh/2 = 4 × 5 ÷ 2 = 10.

4. Two such triangles make a parallelogram.

5. State the result in the form requested: 10 cm².

Answer: 10 cm²

Worked example 2

A triangle has base 14 cm and perpendicular height 4 cm. Select all expressions that give its area in cm².
A. 7 × 4
B. (14 + 4) ÷ 2
C. 14 × 4
D. (14 × 4) ÷ 2

1. Two matching triangles form a parallelogram, so A = (base × perpendicular height)/2.

2. Triangle area is half the matching parallelogram: (14 × 4)/2 = 28 cm².

3. Multiplying half the base by the height gives the same result.

Answer: A, D - 7 × 4; (14 × 4) ÷ 2

Day 106 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Sketch a triangle with base 8 cm and perpendicular height 3 cm. Find its area.

02 / Constructed response

Sketch a triangle with base 16 cm and perpendicular height 5 cm. Find its area.

03 / Constructed response

Sketch a triangle with base 4 cm and perpendicular height 6 cm. Find its area.

04 / Constructed response

Sketch a triangle with base 14 cm and perpendicular height 5 cm. Find its area.

05 / Constructed response

Sketch a triangle with base 8 cm and perpendicular height 2 cm. Find its area.

06 / Constructed response

Sketch a triangle with base 12 cm and perpendicular height 5 cm. Find its area.

07 / Constructed response

Sketch a triangle with base 8 cm and perpendicular height 3/2 cm. Find its area. Explain why the slanted side is not needed.

08 / Multiple select

A triangle has base 8 cm and perpendicular height 6 cm. Select all expressions that give its area in cm².
A. 8 × 6
B. 4 × 6
C. (8 × 6) ÷ 2
D. (8 + 6) ÷ 2

09 / Multiple select

A triangle has base 10 cm and perpendicular height 5 cm. Select all expressions that give its area in cm².
A. (10 + 5) ÷ 2
B. 5 × 5
C. (10 × 5) ÷ 2
D. 10 × 5

10 / Constructed response

Sketch a triangle with base 10 cm and perpendicular height 7/2 cm. Find its area. Explain why the slanted side is not needed.

Day 106 | Unit 08 | Answer key

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1. 12 cm²

A = bh/2 = 8 × 3 ÷ 2 = 12. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

2. 40 cm²

A = bh/2 = 16 × 5 ÷ 2 = 40. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

3. 12 cm²

A = bh/2 = 4 × 6 ÷ 2 = 12. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

4. 35 cm²

A = bh/2 = 14 × 5 ÷ 2 = 35. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

5. 8 cm²

A = bh/2 = 8 × 2 ÷ 2 = 8. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

6. 30 cm²

A = bh/2 = 12 × 5 ÷ 2 = 30. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

7. 6 cm²

A = bh/2 = 8 × 3/2 ÷ 2 = 6. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

8. B, C - 4 × 6; (8 × 6) ÷ 2

Triangle area is half the matching parallelogram: (8 × 6)/2 = 24 cm². Multiplying half the base by the height gives the same result.

6.G.A.1 | Multiple select

9. B, C - 5 × 5; (10 × 5) ÷ 2

Triangle area is half the matching parallelogram: (10 × 5)/2 = 25 cm². Multiplying half the base by the height gives the same result.

6.G.A.1 | Multiple select

10. 35/2 cm²

A = bh/2 = 10 × 7/2 ÷ 2 = 35/2. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response